Star Cores of DICTO's Solids

Polar spikes always fill: the star core treatment of every new convex shape



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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).
Version 3 corrects two cross-references to this series; Version 2 added the link to the app; the results are unchanged.

Version 4 corrects the notes on methods: the names of our computers and private file paths are removed, with small wording fixes. No result changes.

Abstract. In The Truncated Icosahedron Star (#78) DICTO turned a solid's own face pyramids outwards and wrapped the spikes in a hull, the Star Halo. DICTO proposed giving every one of his new shapes the same star core treatment. We do this for the 54 shapes of DICTO Solids (dataset v6) that are cut from their centre into face pyramids, 48 of them convex. With the tips at a fixed multiple of the face centres, as for TI, the space between the star and its halo fills with a corner piece and an edge piece only for 22 of the 48; for 26 the star sticks out of its halo or leaves a layer those pieces cannot fill. The main theorem gives the rule that always works: put each spike's tip at the pole of its face plane in a sphere larger than the solid. Then the halo is the solid's polar, and the layer between star and halo is tiled exactly by one pyramid over each corner and one tetrahedron over each edge, for every convex solid with its centre inside. At the midsphere the edge tetrahedra flatten into the rhombi of Conway's join; the theorem is the thick form of the join. It is proved, and checked numerically on all 48 shapes and exactly on ten. DICTO's sloping bases, pyramids over each corner whose bases slope to meet at the tips, close every shape tested: they fill the halo of the OCD Stella, the Belted Triacontahedron and the OTT, and give each of nine Queens' slices a skin of its own. Building a second and third star on the first, with DICTO's rule of shorter spikes in the valleys, every run ends without a clash; the symmetric shapes freeze into finished stars, while the eccentric pieces keep growing. A card for every shape closes the paper.

1. Introduction

Cut a solid from its centre into pyramids, one on every face, and turn them outwards. The solid becomes a star. Wrap the star in the tightest skin and a new, larger solid appears: the star's halo.

The truncated icosahedron (TI) was the first solid to get this treatment in full (#78): its 32 face pyramids turned outwards make the TI Star, the hull of their tips is the Star Halo, and the space between them fills with tetrahedra of three kinds. DICTO's Cleo star (#35), the icosahedron with its twenty Cleo-Icos turned out, came before it. Taking the hull of the tips undoes DICTO's growth rule T of Universal PiTet (#38), T(S) = {x : v·x ≤ v·v for every corner v of S}, so every star step is a step of T read backwards (#78, Section 6).

DICTO proposed the same treatment for all his new shapes: "all new shapes given the star core treatment". This paper reports it for the 54 shapes of the data set DICTO Solids: Exact Geometry (v6) that can be cut from their centre into face pyramids. It gives one card per shape (Section 8), and one theorem that explains which rule for the spikes always works (Section 3). Paper B of this series (in preparation) follows the cores of the turns, and #80 the exact growth of the chains.

2. The treatment

Let S be a solid with its centre O inside every face's pyramid, so that S is the union of the pyramids from O over its faces. The treatment has six steps:

1. **Pyramids.** Cut S from O into one pyramid per face.
2. **Spikes.** Turn each pyramid outwards across its face, its tip moved out from the face. In the *centre rule* of #78 the tip of face f is $t \cdot c_f$, with c_f the foot of the perpendicular from O to the face and t = 2 (the pyramid mirrored in its face) or $t = \varphi$ (the golden spike).
3. **Star and halo.** The star is S with its spikes; the halo is the convex hull of the tips. The layer between them is checked for an exact fill by the pieces of #78: over each corner a piece to the tips of its faces, over each edge a tetrahedron to the tips of its two faces, with no overlaps.
4. **Core.** The faces are grouped into families (same number of sides, same distance from O); the core is the intersection of the hulls of each family's tips (#78, Proposition 5).
5. **The core's own turn.** The core gets the same treatment, and so on (paper B).
6. **Chain.** The halo gets the treatment in its turn: the star chain, which is the orbit of T read backwards (#80).

Every shape falls into one of four groups under the centre rule: **A**, the star fills its halo and the core comes back to the shape; **B**, the star fills its halo, the core does not come back to the shape itself; **C**, the star lies inside its halo but the corner and edge pieces do not fill the layer; **D**, the star sticks out of its halo.

3. Main theorem: polar spikes always fill

Put every spike's tip not at a multiple of the face centre but at the face's partner point on the far side of a large sphere. Then the gaps between the spikes always fill, for every convex solid, with only two kinds of piece.

For a face plane $n \cdot x = c$ ($c > 0$) and a radius ρ, the *pole* of the plane in the sphere of radius ρ about O is $p = \rho^2 n/c$ (for a unit normal n, the point on the face's axis at distance ρ^2/c from O). Taking poles of all face planes is polar reciprocation, and the hull of the poles is the polar solid $S^* = \{x : v \cdot x \le \rho^2 \text{ for every corner } v\}$.

> **Theorem (polar-spike fill).** Let S be a convex solid with O in its interior, R its circumradius and $\rho > R$. Give every face f the spike with tip p_f, the pole of its plane. Then:
>
> (a) the halo, the hull of the tips, is the polar solid S*; its facet over each corner v lies in the plane $v \cdot x = \rho^2$ and has as corners the tips of the faces round v;
>
> (b) the layer between the star and the halo is tiled exactly, with no gaps and no overlaps, by one *corner pyramid* over each corner v (apex v, base the facet of S* over v) and one *edge tetrahedron* over each edge ab (corners a, b and the tips of the two faces at ab).

> So the pieces number V + E (corners plus edges), and the halo is the star plus V + E pieces.

Proof. (a) is the classical description of the polar: S* is bounded by the planes $v \cdot x = \rho^2$, one per corner v of S, and its corners are the poles of the face planes of S; the corners on the facet of v are the poles of the faces through v. (b) Every piece has positive volume. A spike's tip lies beyond its face because its distance ρ^2/c from O exceeds c ($c \le R < \rho$). A corner pyramid's apex v lies on the inner side of its base plane because $v \cdot v \le R^2 < \rho^2$. For an edge tetrahedron, project along the edge ab: the line of ab becomes a point at distance d from O, the two face planes become lines through it, and the two tips the poles of those lines; a two-line calculation shows the tetrahedron's orientation is the sign of $\rho^2 - d^2$, and $d \le R < \rho$ because the edge's nearest point to O is no farther than its ends. The pieces fit face to face. Each spike side (a, b, p_f) is shared with the edge tetrahedron of ab; each triangle (v, p_f, p_g), for faces f and g meeting at the edge from v, is shared by the corner pyramid of v and the edge tetrahedron of that edge; the faces left over are the spike sides' partners and the facets of S*. So the pieces, all of positive orientation, form a complex whose boundary is the boundary of S* together with the star's surface, and a complex of positively oriented pieces with that boundary covers each point between star and halo exactly once. ∎

The join as the thin limit. If S has a midsphere (every edge touching one sphere of radius ρ_m), then at $\rho = \rho_m$ each edge has $d = \rho$, so every edge tetrahedron is flat: the edge of S and its dual edge cross, and the halo's faces become the rhombi of the join, the hull of S and its polar (Conway's jS). The theorem is the thick version of the join, valid for every convex solid and every $\rho > R$. For ρ between the midsphere and R some pieces would turn inside out, which is why the bound $\rho > R$ is needed.

Checks. The fill was computed for all 48 convex shapes at $\rho^2 = \varphi R^2$ and $2R^2$ (float64, overlaps by the separating-axis test): every layer adds up to the halo minus the star to within 3×10^{-15} of its volume and no two pieces overlap; the layer pieces number from 10 (the Dodeca-13 Needle: 4 corners, 6 edges) to 690 (DPHP: 270 corners, 420 edges). For the ten solids of #80 (GTI, FPD, Palintriacontahedron, Icosaxes, Unborn Axes, Metaxyhexecontahedron, Tweedledum, Samson, David, Delilah) every piece was also checked exactly, in each solid's own number field: nonzero volume, one orientation per kind, and the exact sum equal to the halo's volume.

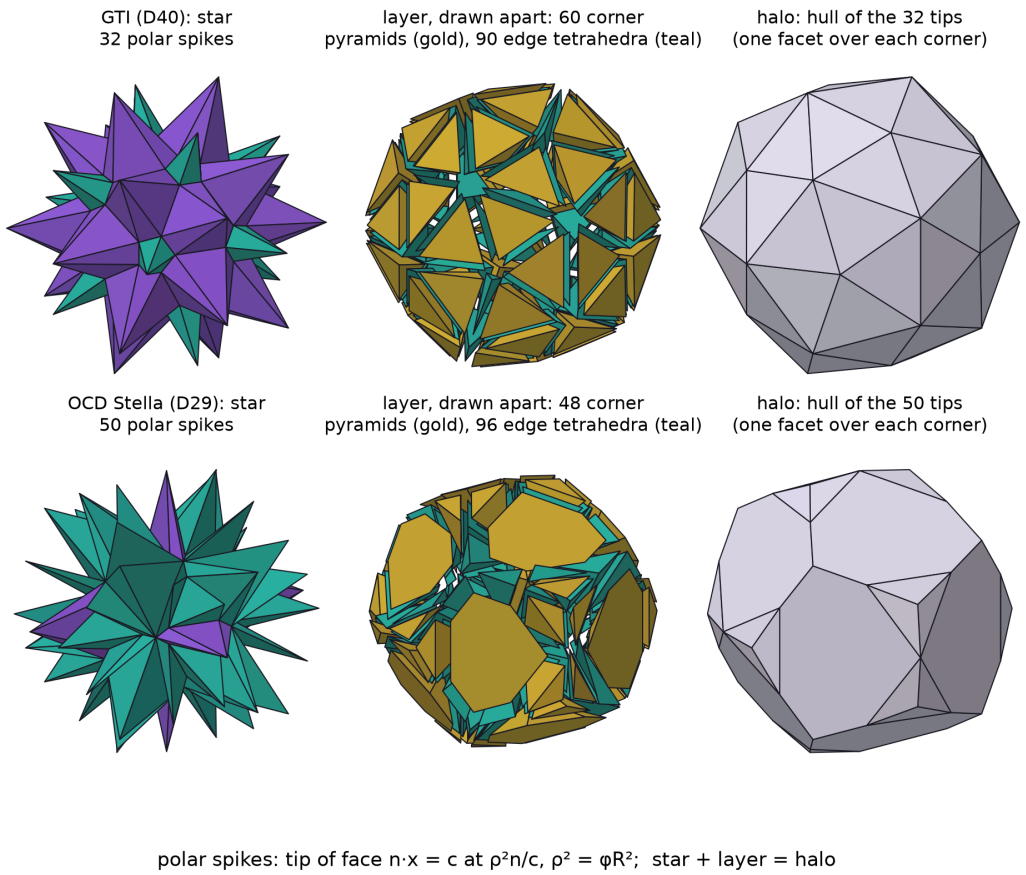


Figure 1. The polar-spike fill at $\rho^2 = \varphi R^2$. Top: the DICTO GTI (D40): its star with 32 polar spikes; the layer drawn apart, 60 corner pyramids (gold) and 90 edge tetrahedra (teal); the halo, the hull of the 32 tips, with one facet over each corner. Bottom: the same for the OCD Stella (D29): 50 spikes, 48 corner pyramids, 96 edge tetrahedra. In each row the star and the layer add up to the halo.

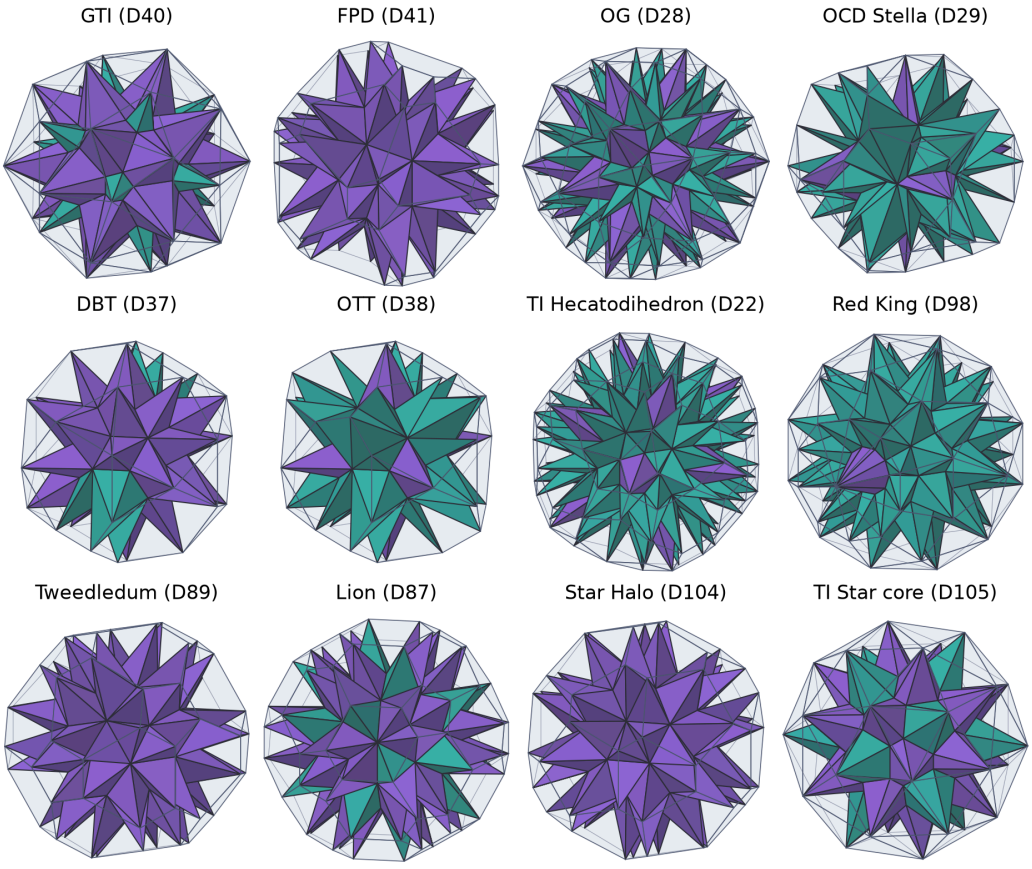


Figure 2. Twelve of DICTO's shapes with polar spikes ($\rho^2 = \varphi R^2$) inside their see-through halos. Spikes on the largest family of faces violet, the others teal.

4. The survey: what the centre rule gives

With the tips at a fixed multiple of the face centres, the trick of the TI Star works for fewer than half of the shapes. The polar rule mends all the rest.

Table 1 sorts the 54 shapes by what the centre rule gives at $t = \varphi$ ($t = 2$ gives the same groups). Only 22 of the 48 convex shapes behave like TI. For the 26 others the tips at a fixed multiple of the face centres are wrong for the shape: where a solid's faces lie at very different distances from the centre, the near faces' tips fall short and the far faces' tips overshoot, so the star either sticks out of the hull of its own tips (group D) or leaves gaps round its high corners that the corner and edge pieces cannot reach (group C). The polar rule of the theorem has no such failure: with it every one of the 48 convex shapes fills, at both $\rho^2 = \varphi R^2$ and $\rho^2 = 2R^2$, whichever group the centre rule put it in. For the faces that all lie at one distance from the centre (TI and the face-transitive shapes) the two rules give the same tips, up to scale.

Group	Centre rule (t = φ)	Convex	Non-convex	Convex shapes
A	star fills, core comes back to the shape	2	0	GTI (golden truncated icosahedron), TI Star core
B	star fills, core does not come back to the shape	20	0	Truncated-Icosahedral Hecatodihedron, Ogdoecontahedron (OG), Metaxyhexecontahedron, Palintriacontahedron, FPD (flat pentakis dodecahedron), KRE, PHE, TPP, DPHP, TPH, TQH, TQP, TQHD, Lion, Jabberwock, Tweedledum, Tweedledee, Jubjub, Red King, Star Halo
C	star inside halo, layer not filled by corner and edge pieces	12	5	OCD Stella, Belted Triacontahedron (DBT), OTT (Octahedrally Truncated Triacontahedron), Mad Hatter, March Hare, Alice, White Rabbit, Haigha, Hatta, Walrus, Red Knight, White Knight
D	star sticks out of its halo	14	1	skewed rhombic dodecahedron, skewed elongated dodecahedron ED F16, skewed elongated dodecahedron ED F18, Dodeca-13 Wedge, Dodeca-13 Needle, Samson, Sonsam, Lahdeli, Ian Dury, Blockhead, Chucky, Fang, Triskelion Tetrahedron, Ganesha

Table 1. The 54 shapes of DICTO Solids v6 cut from their centre into face pyramids, by group under the centre rule. The non-convex six (the DICTO Jewel, Lotus Jewel Axes, Dodecaxes, Keyaxes, Kagemusha and the Golden TI Star) are left to a later paper: a pole needs a face plane that misses the centre, which holds, but the layer pieces can then cut into the solid.

5. DICTO's sloping bases

DICTO proposed keeping the centre-rule tips and filling the gap round each high corner with pyramids taken in order round the corner, their bases sloping to meet the tips.

For the twelve shapes with high-degree corners that the centre rule leaves unfilled (group C) and that DICTO named, we tested his sloping bases at t = φ and t = 2: the centre-rule tips are kept, and round each corner a ring of pyramids from the corner is laid in order, each with a triangular base on three tips, so that neighbouring pyramids meet along the lines between tips; edge tetrahedra close the gaps over the edges. No two pieces overlap in any of the 24 runs. For the OCD Stella, the Belted Triacontahedron and the OTT the pieces fill the convex hull of the tips exactly. For the nine Queens' slices of #76 they close up into a solid of their own, slightly inside the hull (0.2 to 4 per cent less in volume): a new outer solid for each slice, its faces the sloping bases.

D	Shape	Corner pyramids	Edge tetrahedra	Star + pieces / solid	Hull of tips / solid	Result
D29	OCD Stella	96	96	3.1922	3.1922	fills the halo
D37	Belted Triacontahedron (DBT)	60	60	2.9682	2.9682	fills the halo
D38	OTT (Octahedrally Truncated Triacontahedron)	72	60	3.0877	3.0877	fills the halo
D91	Mad Hatter	432	360	4.0883	4.0959	own skin, inside the halo
D92	March Hare	432	360	4.0929	4.1065	own skin, inside the halo
D94	Alice	432	360	4.0807	4.0835	own skin, inside the halo
D95	White Rabbit	432	360	4.1028	4.1154	own skin, inside the halo
D96	Haigha	114	171	3.5644	3.6041	own skin, inside the halo
D97	Hatta	114	171	3.5346	3.6215	own skin, inside the halo
D99	Walrus	120	180	3.5368	3.6992	own skin, inside the halo
D100	Red Knight	120	180	3.5547	3.6727	own skin, inside the halo
D101	White Knight	120	180	3.5838	3.6794	own skin, inside the halo

Table 2. DICTO's sloping bases at t = φ (t = 2 gives the same results, with other volumes). Volumes as multiples of the solid's; float64, overlaps by the separating-axis test.

6. The cores

Take the hull of each family's tips and keep only what lies inside all of them: that is the core of the turn. Turn after turn, the cores of most shapes settle into a pair that swaps back and forth.

The core of a turn (#78, Proposition 5) was followed exactly, turn after turn, for all 48 convex shapes (with the tips at the feet of the face perpendiculars; a factor t only scales the core). Every one of the 29 shapes whose cores keep forming settles into a **cycle of period two**: 2 at once (the GTI and the TI Star core, whose cores bring back the shape itself), 22 after one turn and 5 after two. The other 19 reach a turn where no family of tips spans a solid (every family has fewer than four tips or lies in one plane; for Fang the centre is not strictly inside one family's hull): the low-symmetry shapes at the first turn, seven Queens' slices at the second and Tweedledee at the third. The two-turn scale in radius is $t^2 \cos^2\theta$, with θ the angle between the two kinds of symmetry axis carrying the cores' faces: $t^2(5 + 2\sqrt{5})/15$ for the icosahedral shapes, $t^2/3$ for the octahedral (OCD Stella, OTT, Lahdeli) and $t^2/9$ for the tetrahedral (Samson, the Triskelion Tetrahedron, Alice, the White Rabbit). The Belted Triacontahedron alone has another value, because only one family of its cores is solid at each turn. Paper B proves the period two for cores with two families on two kinds of axis and studies the exceptions.

An earlier float survey reported a returning core only for the GTI and the TI Star core: it asked whether the core of the core is the shape itself, which is the case of a cycle entered at once. The exact run (study 2b) confirms both and finds the cycles entered later.

7. Building on the star

DICTO proposed taking the star itself as the next solid and spiking its triangles in turn. Done naively, the new spikes clash in the valleys; with DICTO's rule of shorter spikes in the valleys, the symmetric shapes stop growing and become finished stars.

Taking the star itself as the next solid, its triangles are spiked in turn. With the tips over the triangles' centres (the centre rule), or over the nearest corner of the star below (a local rule), the new spikes of neighbouring triangles cross in the valleys between the old spikes at the second step, for every shape tested. DICTO's valley rule shortens each spike just enough to clear its neighbours: the ridges keep full spikes and the valleys flatten. With it all 100 runs (50 shapes, t = φ and t = 2, at most 200 faces to start and 2,500 triangles in all) end with no clash, and all but one reach the third step (DPHP at t = φ reaches the cap after two).

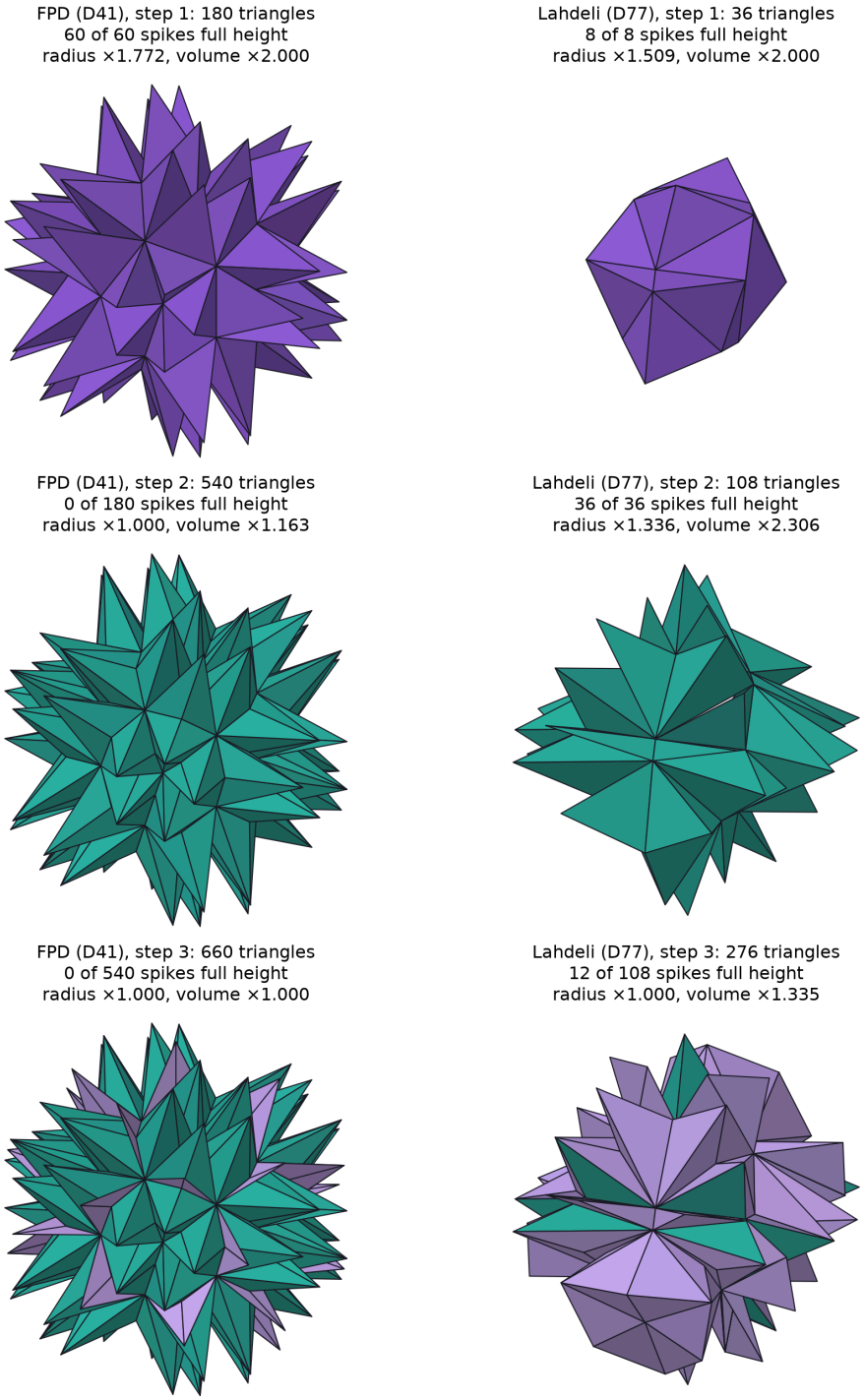


Figure 3. DICTO's valley rule at t = 2, steps one to three; each spike is shortened until it clears the pieces already built and its neighbours. Left: the FPD (D41) stops growing: its radius stays the same from step two and its volume from step three. Right: Lahdeli (D77), an eccentric piece, still adds a third of its volume at step three, although its radius no longer grows. Columns: the FPD, Lahdeli; rows: steps one to three. Spikes of step one violet, step two teal, step three lilac.

The symmetric shapes freeze. At the third step they add less than 0.1 per cent of volume: they are finished stars. At t = φ, 16 shapes freeze (OCD Stella, Palintriacontahedron, Belted Triacontahedron (DBT), OTT (Octahedrally Truncated Triacontahedron), FPD (flat pentakis dodecahedron), PHE, TPP, TPH, TQH, TQHD, Lion, Jabberwock, Tweedledum, Red Knight, Golden TI Star, Star Halo); at t = 2, 27. **The eccentric pieces keep growing**, in volume (their radius levels off by the third step). Over steps two and three at t = 2 the largest growth in volume is Lahdeli ×3.08; Chucky ×2.69; Triskelion Tetrahedron ×2.67; Dodeca-13 Wedge ×2.54; Samson ×2.47; Jewel (EKP windows; first called Dragon Jewel) ×2.46; skewed elongated dodecahedron ED F16 ×2.41. These are the pieces DICTO built as parts rather than as whole symmetric solids, and they are the subject of a later paper of this series.

8. The cards

One line per convex shape. Halo: the volume of the polar-spike halo at $\rho^2 = \varphi R^2$ over the shape's. Pieces: corner pyramids plus edge tetrahedra of the polar fill (V + E). Core: the exact core cycle (paper B). T-chain: the star chain read as DICTO's rule T (#80). Valley: growth over steps two and three under DICTO's valley rule, or 'freezes'.

D	Shape	Sym.	V	F	Group	Halo	Pieces	Core	T-chain	Valley φ	Valley ×2
D1	skewed rhombic dodecahedron	C2h	14	12	D	40.997	38	no core at turn 1	open (8 steps)	×1.21	×1.94
D2	skewed elongated dodecahedron ED F16	C2h	18	12	D	93.397	46	no core at turn 1	–	×1.13	×2.41
D3	skewed elongated dodecahedron ED F18	C2h	18	12	D	47.275	46	no core at turn 1	–	×1.18	×1.91
D6	Dodeca-13 Wedge	C2v	8	6	D	892.626	20	no core at turn 1	no centre	×1.73	×2.54
D7	Dodeca-13 Needle	C3v	4	4	D	296388.001	10	no core at turn 1	no centre	×1.51	×2.17
D22	Truncated-Icosahedral Hecatodihedron	Ih	80	102	B	5.806	260	period 2 from turn 1	–	×1.04	freezes
D28	Ogdoecontahedron (OG)	Ih	102	80	B	5.980	282	period 2 from turn 1	open (8 steps)	×1.03	freezes
D29	OCD Stella	Th	48	50	C	8.649	144	period 2 from turn 1	open (8 steps)	freezes	freezes
D30	Metaxyhexecontahedron	Ih	32	60	B	6.769	122	period 2 from turn 1	period 4	×1.29	freezes
D31	Palintriacontahedron	Ih	32	60	B	6.572	122	period 2 from turn 1	period 4	freezes	freezes
D37	Belted Triacontahedron (DBT)	D3d	30	32	C	7.870	90	period 2 from turn 2	open (8 steps)	freezes	freezes
D38	OTT (Octahedrally Truncated Triacontahedron)	Th	24	38	C	8.299	84	period 2 from turn 1	open (8 steps)	freezes	freezes
D40	GTI (golden truncated icosahedron)	Ih	60	32	A	5.635	150	period 2 at once	period 4	×1.61	freezes
D41	FPD (flat pentakis dodecahedron)	Ih	32	60	B	6.572	122	period 2 from turn 1	period 4	freezes	freezes
D45	KRE	Ih	92	90	B	5.582	272	period 2 from turn 2	–	×1.04	freezes
D46	PHE	Ih	180	92	B	4.982	450	period 2 from turn 1	–	freezes	×1.08

D	Shape	Sym.	V	F	Group	Halo	Pieces	Core	T-chain	Valley φ	Valley ×2
D47	TPP	Ih	212	180	B	5.034	602	period 2 from turn 2	–	freezes	freezes
D48	DPHP	Ih	270	152	B	4.594	690	period 2 from turn 1	–	×1.10	freezes
D49	TPH	Ih	80	132	B	5.033	290	period 2 from turn 1	open (8 steps)	freezes	freezes
D50	TQH	Ih	132	140	B	5.045	402	period 2 from turn 1	open (8 steps)	freezes	freezes
D51	TQP	Ih	140	162	B	5.762	440	period 2 from turn 1	open (8 steps)	×1.04	freezes
D52	TQHD	Ih	162	200	B	4.807	522	period 2 from turn 1	open (8 steps)	freezes	freezes
D71	Samson	Td	16	10	D	44.659	40	period 2 from turn 1	period 4	×1.19	×2.47
D76	Sonsam	C3v	19	13	D	50.231	49	no core at turn 1	no centre	×1.12	×2.21
D77	Lahdeli	D3d	12	8	D	21.660	30	period 2 from turn 2	open (8 steps)	×1.22	×3.08
D78	Ian Dury	C3v	16	13	D	26.982	43	no core at turn 1	no centre	×1.23	×2.10
D79	Blockhead	C3v	10	10	D	20.254	28	no core at turn 1	no centre	×1.36	×2.33
D80	Chucky	C3v	19	13	D	23.692	49	no core at turn 1	no centre	×1.59	×2.69
D81	Fang	C3v	7	7	D	171.421	19	no core at turn 1	no centre	×1.56	×2.37
D82	Triskelion Tetrahedron	Td	16	10	D	22.337	40	period 2 from turn 1	open (8 steps)	×1.15	×2.67
D84	Ganesha	C3v	30	17	D	31.860	75	no core at turn 1	no centre	×1.21	×1.83
D87	Lion	Ih	42	80	B	5.578	162	period 2 from turn 2	open (8 steps)	freezes	freezes
D88	Jabberwock	Ih	72	140	B	6.118	282	period 2 from turn 1	open (8 steps)	freezes	freezes
D89	Tweedledum	Ih	32	60	B	6.284	122	period 2 from turn 1	period 4	freezes	freezes
D90	Tweedledee	D5h	32	60	B	7.926	122	no core at turn 3	no centre	×1.42	freezes
D91	Mad Hatter	D3h	144	218	C	5.366	504	no core at turn 2	no centre	–	–
D92	March Hare	D3h	144	218	C	5.522	504	no core at turn 2	no centre	–	–
D93	Jubjub	D3h	30	56	B	9.760	114	no core at turn 2	no centre	×1.33	freezes
D94	Alice	Td	144	218	C	5.260	504	period 2 from turn 1	open (8 steps)	–	–
D95	White Rabbit	Td	144	218	C	5.469	504	period 2 from turn 1	open (8 steps)	–	–
D96	Haigha	D3h	114	59	C	4.982	285	no core at turn 2	no centre	×1.31	×1.18
D97	Hatta	D3h	114	59	C	5.027	285	no core at turn 2	no centre	×1.31	×1.20
D98	Red King	D10h	80	52	B	5.106	210	no core at turn 2	open (8 steps)	×1.35	freezes

D	Shape	Sym.	V	F	Group	Halo	Pieces	Core	T-chain	Valley φ	Valley ×2
D99	Walrus	D5h	120	62	C	5.006	300	no core at turn 2	no centre	×1.34	×1.20
D100	Red Knight	Ih	120	62	C	4.909	300	period 2 from turn 1	open (8 steps)	freezes	freezes
D101	White Knight	Ih	120	62	C	4.818	300	period 2 from turn 1	open (8 steps)	×1.30	freezes
D104	Star Halo	Ih	32	60	B	6.175	122	period 2 from turn 1	–	freezes	freezes
D105	TI Star core	Ih	60	32	A	5.315	150	period 2 at once	–	×1.66	freezes

Table 3. The cards. Group: the centre rule's group of Table 1. V, F: corners and faces (pieces V + E = 2V + F − 2 by Euler). '–': not run (over 200 faces, or no data).

9. What is new and what is known

- **Known, credited:** DICTO's growth rule T (#38) and the TI Star with its halo and fills (#78); polar reciprocation and the description of the polar by the poles of the face planes; Conway's join jS and the fact that a solid with a midsphere and its polar cross edge to edge, the hull having four-sided faces (G. W. Hart, Conway notation for polyhedra; D. Eppstein, "Triangles and squares", COMB01, Barcelona 2001, slides); the hull and core of a regular polytope and its reciprocal (H. S. M. Coxeter, "Reciprocating the regular polytopes", J. London Math. Soc. 55 (1997) 549–557, abstract read).
- **New, as far as the sources checked show:** the polar-spike fill theorem for every ρ greater than the circumradius, with its count V + E of pieces and the sign $\rho^2 - d^2$ that makes the bound exact; the survey of DICTO's 54 shapes under both rules; DICTO's sloping bases and the skins they give the Queens' slices; DICTO's valley rule and the freezing of the symmetric stars.
- **Not decided here:** the non-convex shapes under the polar rule (a pole needs the face plane to miss O, and the layer pieces may then overlap the solid); the shapes of the Queens' skins; whether the frozen stars are exactly stationary (the runs stop at three steps).

In the app. Polyhedraverse, Star step (polyhedraverse.dictospheres.com/?open=starstep:TRUNCATED_ICOSAHEDRON), gives any convex solid the treatment one tap at a time: the star, the halo with its layer of corner pyramids and edge tetrahedra, then the halo as the next solid.

10. Conclusion

DICTO's star core treatment reaches all 48 of his convex shapes once the spikes follow the polar rule. With the tips at a fixed multiple of the face centres, the rule of the TI Star, fewer than half of the shapes fill; with the tips at the poles of the face planes in any sphere larger than the shape, every one does, with exactly one pyramid per corner and one tetrahedron per edge, and the halo is the shape's polar. This is the thick form of Conway's join, and it gives this series its standard star. DICTO's sloping bases close every shape tested, and his valley rule lets a star be built upon without clashes: the symmetric shapes freeze into finished stars, the eccentric pieces grow on. The cards record each shape's group, halo, pieces, core cycle, chain and valley growth.

Sources checked (30–31 October 2026; full list with every query in study 2026-10-30-star-cores-exact, prior-art-study6.md and README section 3). *Reference library* (ask-library, 22 queries, among them "hull of a polyhedron and its dual / join", "polar spikes and Kleetope", "dissection between a polyhedron and its polar", "Conway notation, join, kis and the midsphere"): no source on the layer between a solid and its polar; nearest O'Rourke 2023 (polar zonohedra, unfolding), Cunningham and Mixer 2016 (abstract polytopes), Messer 1995 (stellation cores). *Web, read:* Wikipedia "Dual polyhedron", "Conway polyhedron notation", "Polytope compound"; G. W. Hart, "Conway notation for polyhedra" (georgehart.com); Antiprism documentation of conway and pol_recip; D. Eppstein, "Triangles and squares", COMB01 slides (2001). *Abstract only:* Coxeter 1997. DICTO's papers #35, #38 and #78 are cited for their results.

Data and code. Study 2026-10-30-star-cores: scripts/star_core.py (the treatment, groups), polar_spikes.py (polar fill), sloping_bases.py (DICTO's sloping bases), star_build.py and star_valley.py (building on the star; valley rule), data/star-cores.json, polar-spikes.json, sloping-bases.json, star-build.json, star-build-local.json, star-valley.json. Study 2026-10-30-star-cores-exact: scripts/study1.py (exact polar fill on ten solids), study2b.py (exact core cycles), data/study1.json, study2b.json. Study 2026-10-30-star-chain-census: survey2.json (T-chains). Shapes from DICTO Solids: Exact Geometry v6 (10.5281/zenodo.23790799).